

Advanced Enterprise Network

Mixed Routing Design

Comprehensive network architecture featuring 5 managed switches, 8 workstations, 3 clients, and 6 routers with dual routing protocols, 3 north routers using static routing and 3 right routers implementing OSPF for optimal performance and reliability

Digital Egypt Pioneer Initiative (DEPI) Graduation Project

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Project Overview

This project details the design and configuration of a multi-department enterprise network featuring a mixed routing architecture, comprehensive VLAN segmentation, and essential network services. The implementation uses Huawei routers and switches within the eNSP simulation environment.

Devices: 6 routers, 5 switches, 8 PCs, 3 clients, 2 APs and 2 servers.

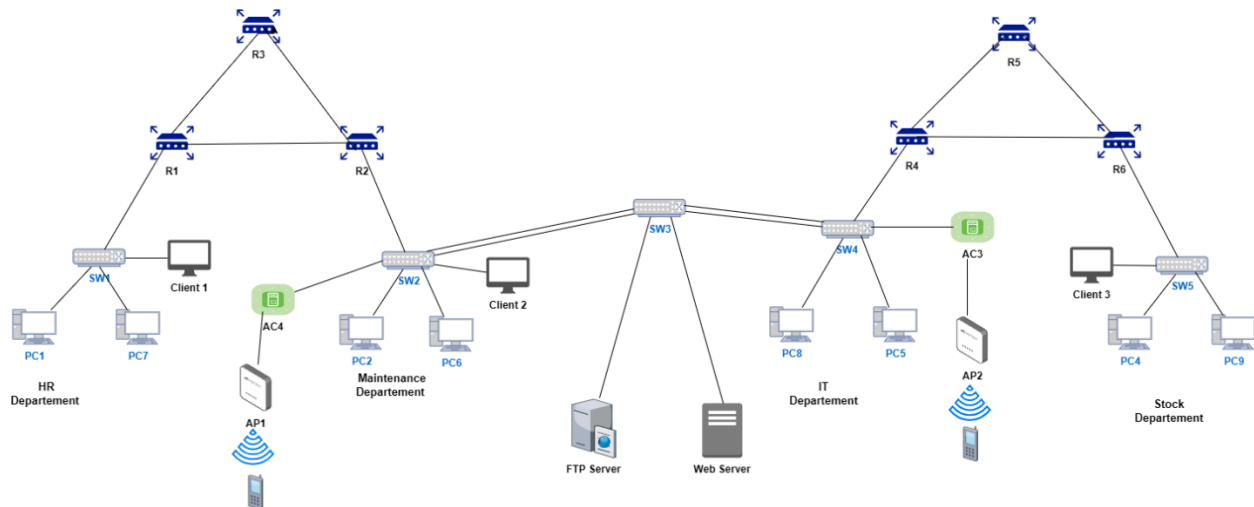
Features implemented: VLANs, static routing, OSPF routing, ACLs, Access Points, DHCP & FTP servers

Key Features Implemented

Feature	Description
Network Architecture	Hierarchical design with 6 routers, 5 switches, 8 PCs, 3 clients, 2 Access Points (APs), and 2 servers.
VLAN Segmentation	Five distinct VLANs (10, 20, 30, 40, 50) for traffic isolation and security.
Mixed Routing	Static Routing for the North Routers (AR1, AR2, AR3) and OSPF for the South Routers (AR4, AR5, AR6) to provide both predictable paths and dynamic adaptability.
First Hop Redundancy	VRRP (Virtual Router Redundancy Protocol) implemented on the Distribution Layer switches (LSW1, LSW2, LSW4, LSW5) to ensure high availability for VLAN gateways.
Network Services	FTP Server, Web Server, and Wireless Access Control (AC/AP).

Network Topology Overview

The network is structured into three main layers: Core (Routers), Distribution (Switches), and Access (End Devices)



- Six routers configured with dual routing protocols for optimal performance
- Three north routers implementing static routing for predictable paths
- Three right routers using OSPF for dynamic adaptive routing
- Five managed switches providing segmented connectivity for eight workstations

This hybrid network design combines the reliability of static routing with the adaptability of OSPF protocols, providing both predictable traffic flows and dynamic network resilience for enterprise operations. The mixed routing approach ensures optimal performance while maintaining network stability and providing automatic failover capabilities for critical business operations.

Network Topology Description

- The network is divided into five VLANs (10, 20, 30, 40, 50).
- There are 6 routers (R1–R6) connected in a hierarchical manner to provide inter-network connectivity.
- 5 switches are used for VLAN segmentation and device access.
- FTP and Web servers are placed in VLAN 50, while users are distributed across VLANs 10–40
- There are 2 access controller and 2 access points providing a wireless communication connection for 2 cell phones.

Device Naming Convention

Device Type	Name in Diagram	Count
Routers (R)	R1, R2, R3, R4, R5, R6	6
Switches (SW)	SW1, SW2, SW3, SW4, SW5	5
Access Controllers (AC)	AC3, AC4	2
Access Points (AP)	AP1, AP2	2
Servers	FTP Server, Web Server	2
End Devices	PC1, PC2, PC4, PC5, PC6, PC7, PC8, PC9, Client1, Client2, Client3, Cellphone1, Cellphone2	13

Department Information

This section outlines the departmental structure of the enterprise network, including VLAN assignments and IP addressing for each department. The network is segmented into four departments, HR, stock, maintenance and IT, to ensure security and efficient traffic management. The HR and Stock departments are isolated from each other using Access Control Lists (ACLs) to prevent unauthorized communication and enhance data security.

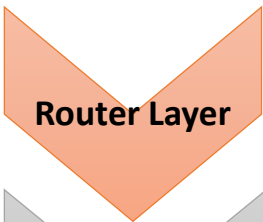
Department	VLAN ID	IP Address Range	PCs	Notes
HR	10	192.168.3.0/24	PC1, PC7, Client1	Isolated from Stock department via ACL.
Maintenance	20	192.168.4.0/24	PC2, PC6, Client2, Cellphone 2	General access to network resources.
IT	30	10.200.1.0/24	PC5, PC8, Cellphone1	Access to development servers.
Stock	40	10.150.1.0/24	PC4, PC9, Client3	Isolated from HR department via ACL.

Server Zone

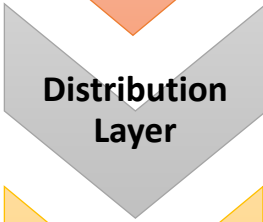
The server zone is a dedicated network segment for hosting critical network services. It is isolated from the user VLANs to protect the servers from unauthorized access and potential security threats.

Server	VLAN ID	IP Address	Gateway
FTP Server	50	192.168.50.100	192.168.50.1
Web Server	50	192.168.50.101	192.168.50.1

Network Hierarchy Architecture



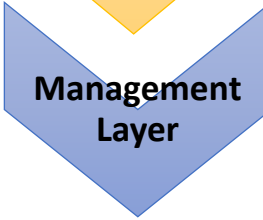
Six routers providing dual routing protocols with three static and three OSPF routers



Five managed switches handling traffic aggregation and VLAN segmentation



Ten workstations distributed across switch infrastructure for optimal connectivity



Centralized monitoring and configuration management for mixed routing protocols

VLAN and IP Addressing Scheme

The network is segmented into five VLANs, each with a dedicated subnet and a VRRP-configured gateway.

VLAN ID	VLAN Name	Network Address	Gateway (VRRP VIP)	Device Assignments
VLAN 10	PC/Client Access	192.168.3.0/24	192.168.3.1	PC1, PC7, Client1
VLAN 20	PC/Client Access	192.168.4.0/24	192.168.4.1	PC2, PC6, Client2, Cellphone 2
VLAN 30	PC Access	10.200.1.0/24	10.200.1.1	PC5, PC8, Cellphone1
VLAN 40	PC Access	10.150.1.0/24	10.150.1.1	PC4, PC9, Client3
VLAN 50	Server Farm	192.168.50.0/24	192.168.50.1	FTP Server, Web Server

Detailed IP Address Assignments

Device	VLAN	IP Address	Gateway
PC 1	10	192.168.3.100	192.168.3.1
PC 7	10	192.168.3.101	192.168.3.1
Client 1	10	192.168.3.102	192.168.3.1
PC 2	20	192.168.4.100	192.168.4.1
PC 6	20	192.168.4.101	192.168.4.1
Client 2	20	192.168.4.102	192.168.4.1
PC 5	30	10.200.1.100	10.200.1.1
PC 8	30	10.200.1.101	10.200.1.1
Client 3	40	10.150.1.102	10.150.1.1
PC 4	40	10.150.1.100	10.150.1.1
PC 9	40	10.150.1.101	10.150.1.1
FTP Server	50	192.168.50.100	192.168.50.1
Web Server	50	192.168.50.101	192.168.50.1

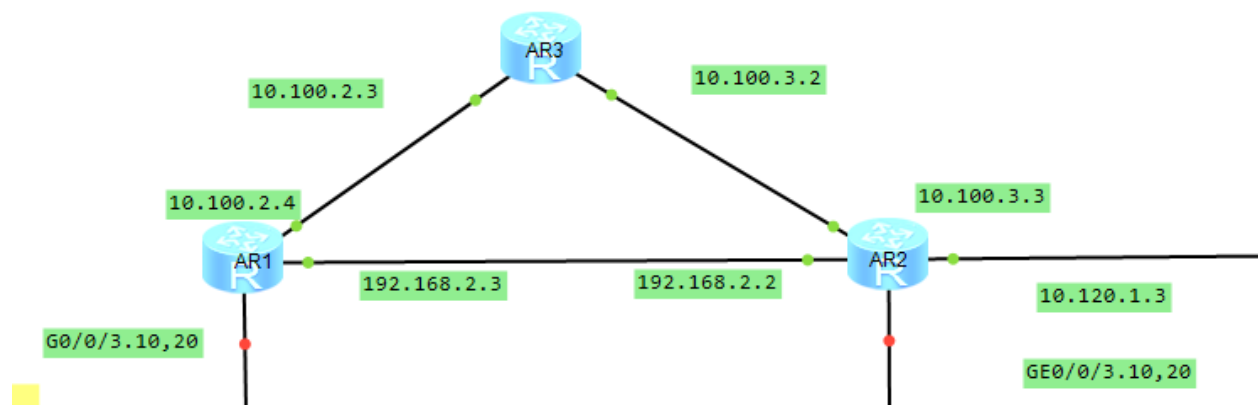
Routing Design

The network employs a Mixed Routing Design to leverage the strengths of both static and dynamic routing protocols.

North Routers (R1, R2, R3) - Static Routing Domain

These routers form the core of the network and use Static Routing for predictable, deterministic traffic paths. This is ideal for the backbone where route changes are infrequent and control over traffic flow is paramount.

Router	IP Address	Connected Routers
R1	10.100.2.4	R3
	192.168.2.3	R2
R2	192.168.2.2	R1
	10.100.3.3	R3
R3	10.100.3.2	R2
	10.100.2.3	R1

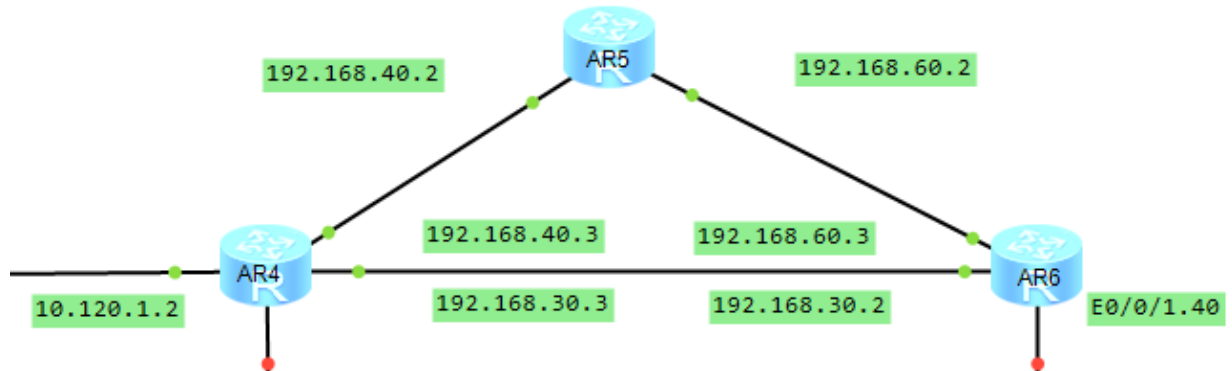


South Routers (AR4, AR5, AR6) - OSPF Routing Domain

These routers connect the Distribution Layer switches and use OSPF (Open Shortest Path First) for dynamic, adaptive routing. OSPF provides automatic topology discovery, fast convergence, and load balancing, which is crucial for the access and distribution layers where network changes and failures are more likely.

Router	IP Address	Connected Routers
R4	192.168.30.3	R6
	192.168.40.3	R5
R5	192.168.40.2	R4

	192.168.60.2	R6
R6	192.168.60.3	R5
	192.168.30.2	R4



Routing Integration

The two routing domains are connected via the Distribution Layer switches and their respective uplink interfaces on the routers. This strategic combination ensures optimal traffic distribution while maintaining network stability through redundant pathways and protocol-specific advantages.

First Hop Redundancy (VRRP) Configuration

VRRP is configured on the VLAN Interfaces (Vlanif) of the Distribution Layer switches to provide a highly available default gateway for all end devices. The VRRP Virtual IP (VIP) is the default gateway for the corresponding VLAN.

VRRP Master/Backup Assignment

The VRRP configuration from the draft document suggests a primary/secondary setup, likely across SW1/SW2 and SW4/SW5, with one switch acting as the master (higher priority) and the other as the backup for each VLAN.

SW2	Vlanif 10 ip address 192.168.3.4 vrrp vrid 10
	Vlanif 20 ip address 192.168.4.4

	vrrp vrid 20
	Vlanif 30 ip address 10.200.1.4 vrrp vrid 30
	Vlanif 40 ip address 10.150.1.4 vrrp vrid 40
	Vlanif 50 ip address 192.168.50.4 vrrp vrid 50
SW3	Vlanif 10 ip address 192.168.3.2 vrrp vrid 10 priority 120
	Vlanif 20 ip address 192.168.4.2 vrrp vrid 20 priority 120
	Vlanif 30 ip address 10.200.1.2 vrrp vrid 30 priority 120
	Vlanif 40 ip address 10.150.1.2 vrrp vrid 40 priority 120
	Vlanif 50 ip address 192.168.50.2 vrrp vrid 50 priority 120
SW4	Vlanif 10 ip address 192.168.3.3 vrrp vrid 10
	Vlanif 20 ip address 192.168.4.3 vrrp vrid 20 priority 120
	Vlanif 30 ip address 10.200.1.3 vrrp vrid 30
	Vlanif 40 ip address 10.150.1.3 vrrp vrid 40
	Vlanif 50 ip address 192.168.50.3

	vrrp vrid 50
SW5	Vlanif 10 ip address 192.168.3.5 vrrp vrid 10
	Vlanif 20 ip address 192.168.4.5 vrrp vrid 20
	Vlanif 30 ip address 10.200.1.5 vrrp vrid 30
	Vlanif 40 ip address 10.150.1.5 vrrp vrid 40
	Vlanif 50 ip address 192.168.50.5 vrrp vrid 50

Security Framework Implementation

1. Access Control Lists (ACLs)

ACLs are implemented to control traffic flow and enforce security policies between different VLANs and network segments.

- **Inter-VLAN Security:** ACLs are applied to the VLAN interfaces on the Distribution Layer switches to filter traffic between user VLANs (10, 20, 30, 40) and the Server Farm (VLAN 50).
- **Protocol Security:** ACLs are used to restrict access to management protocols (e.g., Telnet, SSH) and to filter unauthorized routing updates.

ACL Configuration

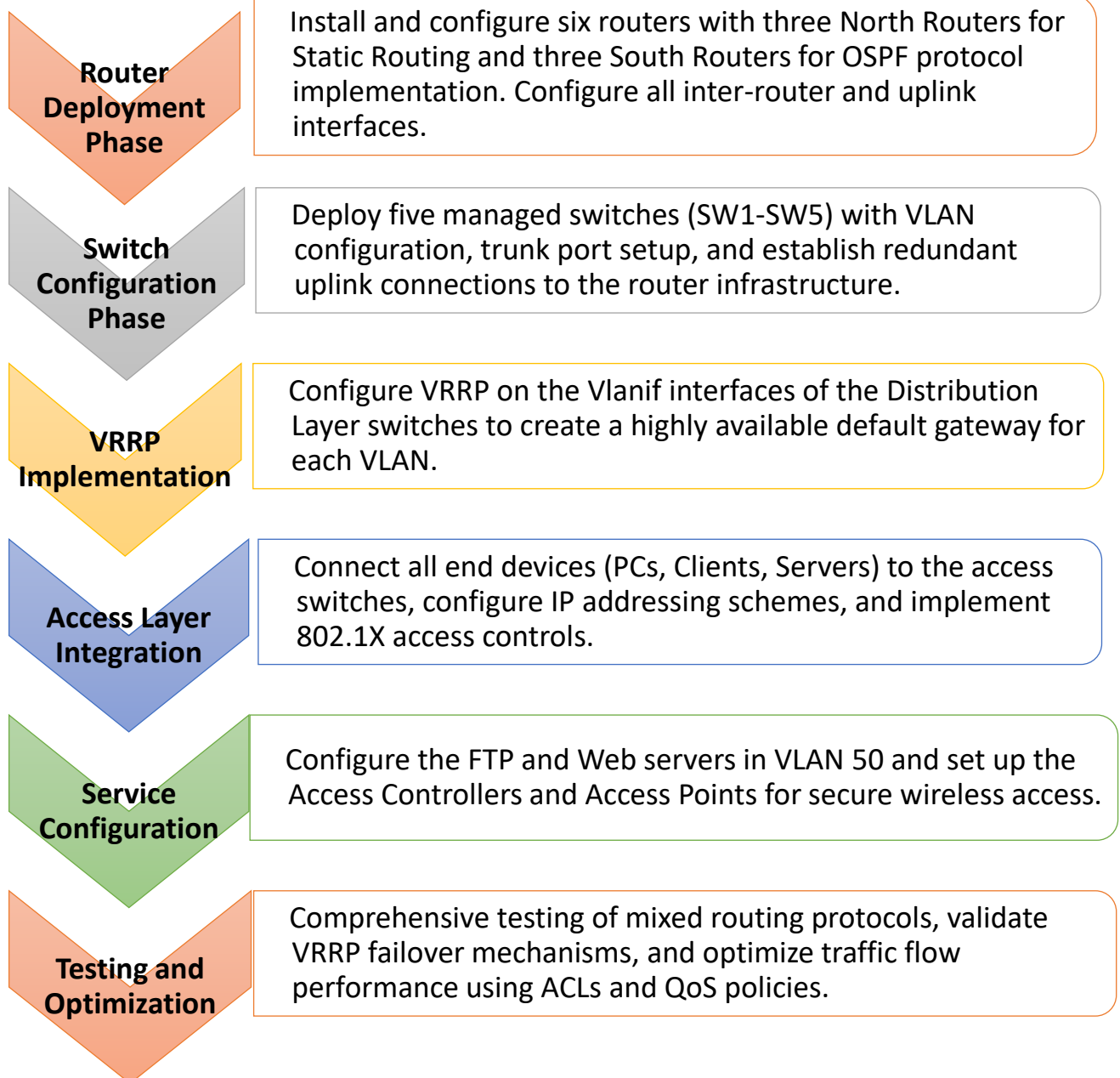
- **Client 1** can access the web server and can also access the FTP server.
- **Client 2** can access the web server but cannot access the FTP server.
- **Client 3** can access the FTP server but cannot access the web server.

2. Wireless Security

The network includes two Access Controllers (AC3, AC4) and two Access Points (AP1, AP2) to provide secure wireless connectivity for mobile devices (Cellphone1, Cellphone2).

- **802.1X Authentication:** Port-based 802.1X authentication is implemented on the access switches to ensure only authorized devices can connect to the network.
- **VLAN Assignment:** Wireless clients are dynamically assigned to their respective VLANs (VLAN 20 for Cellphone2, VLAN 30 for Cellphone1) upon successful authentication.

Implementation Roadmap



Devices Configuration:

- Routers

1- R1

```
AR1
The device is running!
<R-1>
<R-1>disp cu
[V200R003C00]
#
 sysname R-1
#
 board add 0/6 24GE
#
 snmp-agent local-engineid 800007DB0300000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 ---- More ----|
```

```

AR1
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 10.100.2.4 255.255.255.0
#
interface GigabitEthernet0/0/1
ip address 192.168.2.3 255.255.255.0
#
interface GigabitEthernet0/0/2
#
interface GigabitEthernet0/0/2.10
dot1q termination vid 10
ip address 192.168.3.254 255.255.255.0
arp broadcast enable
#
interface GigabitEthernet6/0/0
#
interface GigabitEthernet6/0/1
#
interface GigabitEthernet6/0/2
#
---- More ----|
interface NULL0
#
ip route-static 10.100.3.0 255.255.255.0 10.100.2.3
ip route-static 10.200.1.0 255.255.255.0 192.168.2.2
ip route-static 192.168.4.0 255.255.255.0 192.168.2.2
ip route-static 192.168.4.0 255.255.255.0 10.100.2.3
ip route-static 192.168.50.0 255.255.255.0 192.168.2.2
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#

```

We enabled HTTP management. We configured three routed interfaces (10.100.2.4/24, 192.168.2.3/24, and VLAN 10 gateway 192.168.3.254/24). We added static routes to reach remote networks. We created a basic local firewall zone.

2- R2

```
AR1  AR2
<R-2>
<R-2>
<R-2>
<R-2>disp cu
[V200R003C00]
#
 sysname R-2
#
 board add 0/4 4GET
 board add 0/6 24GE
#
 snmp-agent local-engineid 800007DB0300000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
 acl number 3001
 rule 1 deny tcp source 192.168.4.0 0.0.0.255 destination 192.168.50.0 0.0.0.254
 destination-port eq ftp
#
 aaa
 ---- More ----|

AR1  AR2
#
 aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
 firewall zone Local
 priority 15
#
 interface GigabitEthernet0/0/0
 ip address 10.120.1.3 255.255.255.0
#
 interface GigabitEthernet0/0/1
 ip address 192.168.2.2 255.255.255.0
#
 interface GigabitEthernet0/0/2
 ip address 10.100.3.3 255.255.255.0
#
 interface GigabitEthernet4/0/0
#
 interface GigabitEthernet4/0/0.20
 dot1q termination vid 20
 ip address 192.168.4.254 255.255.255.0
 traffic-filter inbound acl 3001
 arp broadcast enable
 ---- More ----|
```

```
AR1  AR2
#
ospf 1 router-id 4.4.4.4
 area 0.0.0.1
  network 10.100.2.0 0.0.0.255
  network 10.100.3.0 0.0.0.255
  network 10.120.1.0 0.0.0.255
  network 10.150.1.0 0.0.0.255
  network 10.200.1.0 0.0.0.255
  network 192.168.2.0 0.0.0.255
  network 192.168.3.0 0.0.0.255
  network 192.168.4.0 0.0.0.255
  network 192.168.30.0 0.0.0.255
  network 192.168.40.0 0.0.0.255
  network 192.168.50.0 0.0.0.255
  network 192.168.60.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 10.120.1.2
ip route-static 10.100.2.0 255.255.255.0 10.100.3.2
ip route-static 10.100.3.0 255.255.255.0 10.100.2.3
ip route-static 192.168.3.0 255.255.255.0 192.168.2.3
ip route-static 192.168.3.0 255.255.255.0 10.100.3.2 preference 80
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
port-group ip
#
---- More ----|
```

We created ACL 3001 to block FTP traffic from network 192.168.4.0 to 192.168.50.0.

We configured multiple routed interfaces, including VLAN 20 (192.168.4.254/24) and VLAN 50 (192.168.50.254/24).

We applied ACL 3001 inbound on VLAN 20.

We enabled OSPF area 0.0.0.1 for all internal networks to advertise routing dynamically.

We added static routes including a default route toward 10.120.1.2

The firewall local zone was also configured.

3- R3

```
AR3
The device is running!

<R-3>
<R-3>disp cu
[V200R003C00]
#
 sysname R-3
#
 board add 0/6 24GE
#
 snmp-agent local-engineid 800007DB0300000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 ---- More ----
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
 firewall zone Local
 priority 15
#
 interface GigabitEthernet0/0/0
 ip address 10.100.2.3 255.255.255.0
#
 interface GigabitEthernet0/0/1
#
 interface GigabitEthernet0/0/2
 ip address 10.100.3.2 255.255.255.0
#
 interface GigabitEthernet6/0/0
#
 interface GigabitEthernet6/0/1
#
```

```

interface NULL0
#
ip route-static 0.0.0.0 0.0.0.0 10.100.3.3
ip route-static 192.168.3.0 255.255.255.0 10.100.2.4
ip route-static 192.168.4.0 255.255.255.0 10.100.3.3
#
user-interface con 0
    authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
<R-3>|

```

We configured routed interfaces for 10.100.2.3/24 and 10.100.3.2/24.
 We added static routes including a default route via 10.100.3.3 and routes to networks 192.168.3.0 and 192.168.4.0.
 We created the Local firewall zone

4- R4

```

AR3  AR4
The device is running!

<R-4>disp cu
[V200R003C00]
#
sysname R-4
#
board add 0/4 4GET
board add 0/6 24GE
#
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
drop illegal-mac alarm
#
vlan batch 30
#
wlan ac-global carrier id other ac id 0
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
---- More ----|

```

AR3

AR4

```
authorization-scheme default
accounting-scheme default
domain default
domain default admin
local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 10.120.1.2 255.255.255.0
#
interface GigabitEthernet0/0/1
ip address 192.168.30.3 255.255.255.0
#
interface GigabitEthernet0/0/2
ip address 192.168.40.3 255.255.255.0
#
interface GigabitEthernet4/0/0
#
interface GigabitEthernet4/0/0.30
dot1q termination vid 30
ip address 10.200.1.254 255.255.255.0
arp broadcast enable
#
interface GigabitEthernet4/0/1
#
interface GigabitEthernet4/0/2
---- More ----
```

AR3

AR4

```
#
interface NULL0
#
ospf 1 router-id 1.1.1.1
area 0.0.0.1
network 10.100.2.0 0.0.0.255
network 10.100.3.0 0.0.0.255
network 10.120.1.0 0.0.0.255
network 10.150.1.0 0.0.0.255
network 10.200.1.0 0.0.0.255
network 192.168.2.0 0.0.0.255
network 192.168.3.0 0.0.0.255
network 192.168.4.0 0.0.0.255
network 192.168.30.0 0.0.0.255
network 192.168.40.0 0.0.0.255
network 192.168.50.0 0.0.0.255
network 192.168.60.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 10.120.1.3
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
<R-4>
<R-4>
```

We created VLAN 30.

We configured routed interfaces for 10.120.1.2/24, 192.168.30.3/24, and 192.168.40.3/24, plus a VLAN 30 sub interface with gateway 10.200.1.254/24.

We enabled OSPF in area 0.0.0.1 to advertise all internal networks.

We added a default static route via 10.120.1.3.

We created the Local firewall zone

5- R5

```
<R-5>disp cu
[V200R003C00]
#
 sysname R-5
#
 snmp-agent local-engineid 800007DB0300000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
 aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
 ---- More ----
```

AR5

```
local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw)%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
ip address 192.168.40.2 255.255.255.0
#
interface GigabitEthernet0/0/2
ip address 192.168.60.2 255.255.255.0
#
interface NULL0
#
ospf 1 router-id 3.3.3.3
area 0.0.0.1
network 192.168.30.0 0.0.0.255
network 192.168.40.0 0.0.0.255
network 192.168.60.0 0.0.0.255
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
---- More ----|
```

We configured routed interfaces for 192.168.40.2/24 and 192.168.60.2/24.
We enabled OSPF in area 0.0.0.1 to advertise networks 192.168.30.0,
192.168.40.0, and 192.168.60.0.
We created the Local firewall zone

6- R6

AR6

The device is running!

```
<R-6>
<R-6>disp cu
[V200R003C00]
#
sysname R-6
#
snmp-agent local-engineid 800007DB03000000000000
snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
drop illegal-mac alarm
#
wlan ac-global carrier id other ac id 0
#
set cpu-usage threshold 80 restore 75
#
acl number 3001
rule 2 deny tcp source 10.150.1.0 0.0.0.255 destination 192.168.50.0 0.0.0.255
destination-port eq www
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
---- More ----|
```

AR6

```
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw)%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/0.40
dot1q termination vid 40
ip address 10.150.1.254 255.255.255.0
traffic-filter inbound acl 3001
arp broadcast enable
#
interface GigabitEthernet0/0/1
ip address 192.168.30.2 255.255.255.0
#
interface GigabitEthernet0/0/2
ip address 192.168.60.3 255.255.255.0
#
interface NULL0
#
ospf 1 router-id 2.2.2.2
area 0.0.0.1
network 10.100.2.0 0.0.0.255
---- More ----|
```

```
AR6
#
ospf 1 router-id 2.2.2.2
 area 0.0.0.1
  network 10.100.2.0 0.0.0.255
  network 10.100.3.0 0.0.0.255
  network 10.120.1.0 0.0.0.255
  network 10.150.1.0 0.0.0.255
  network 10.200.1.0 0.0.0.255
  network 192.168.2.0 0.0.0.255
  network 192.168.3.0 0.0.0.255
  network 192.168.4.0 0.0.0.255
  network 192.168.30.0 0.0.0.255
  network 192.168.40.0 0.0.0.255
  network 192.168.50.0 0.0.0.255
  network 192.168.60.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 192.168.30.3
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
port-group q
#
wlan ac
#
return
```

We created ACL 3001 to deny HTTP traffic from network 10.150.1.0 (VLAN 40, stock department) going to 192.168.50.0 (VLAN 50).

We configured interfaces including VLAN 40 gateway 10.150.1.254/24 with the ACL applied inbound, plus routed interfaces 192.168.30.2/24 and 192.168.60.3/24.

We enabled OSPF in area 0.0.0.1 for all internal networks.

We added a default static route via 192.168.30.3.

We created the Local firewall zone.

- Switches

1- SW1

```
LSW1
<SW1>
<SW1>disp cu
#
sysname SW1
#
vlan batch 10 20 30 50
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
interface Vlanif1
#
interface MEth0/0/1
---- More ----|
```



```
LSW1
interface Vlanif1
#
interface MEth0/0/1
#
interface Ethernet0/0/1
port link-type access
port default vlan 10
#
interface Ethernet0/0/2
port link-type access
port default vlan 10
#
interface Ethernet0/0/3
port link-type access
port default vlan 10
#
interface Ethernet0/0/4
#
interface Ethernet0/0/5
#
interface Ethernet0/0/6
#
interface Ethernet0/0/7
#
interface Ethernet0/0/8
#
interface Ethernet0/0/9
#
interface Ethernet0/0/10
---- More ----
interface GigabitEthernet0/0/1
port link-type access
#
interface GigabitEthernet0/0/2
port link-type trunk
port trunk allow-pass vlan 10 20 50
#
interface NULL0
#
user-interface con 0
user-interface vty 0 4
#
return
```

We created VLANs 10, 20, 30, and 50 on the switch.
We configured access ports (Eth0/0/1–3) to belong to VLAN 10.
We configured GigabitEthernet0/0/2 as a trunk port, allowing VLANs 10,20, and 50.

2- SW2

LSW2

```
<SW2>
<SW2>
<SW2>
<SW2>disp cu
#
sysname SW2
#
undo info-center enable
#
vlan batch 10 20 30 40 50 100
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 ---- More ----
```

LSW2

```
domain default_admin
local-user admin password simple admin
local-user admin service-type http
#
interface Vlanif1
#
interface Vlanif10
 ip address 192.168.3.4 255.255.255.0
 vrrp vrid 10 virtual-ip 192.168.3.1
#
interface Vlanif20
 ip address 192.168.4.4 255.255.255.0
 vrrp vrid 20 virtual-ip 192.168.4.1
#
interface Vlanif30
 ip address 10.200.1.4 255.255.255.0
 vrrp vrid 30 virtual-ip 10.200.1.1
#
interface Vlanif40
 ip address 10.150.1.4 255.255.255.0
 vrrp vrid 40 virtual-ip 10.150.1.1
#
interface Vlanif50
 ip address 192.168.50.4 255.255.255.0
 vrrp vrid 50 virtual-ip 192.168.50.1
#
interface MEth0/0/1
#
interface Ethernet0/0/1
 ---- More ----
```

LSW2

```
#
interface MEth0/0/1
#
interface Ethernet0/0/1
port link-type access
port default vlan 20
#
interface Ethernet0/0/2
port link-type access
port default vlan 20
#
interface Ethernet0/0/3
port link-type trunk
port trunk allow-pass vlan 10 20 50 100
#
interface Ethernet0/0/4
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50
#
interface Ethernet0/0/5
port link-type access
port default vlan 20
#
interface Ethernet0/0/6
#
interface Ethernet0/0/7
#
interface Ethernet0/0/8
#
---- More ----
interface GigabitEthernet0/0/1
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50
#
interface GigabitEthernet0/0/2
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50
#
interface NULL0
#
ip route-static 192.168.3.0 255.255.255.0 192.168.2.3
#
user-interface con 0
user-interface vty 0 4
#
return
```

We created VLANs 10, 20, 30, 40, and 50

We configured Layer 3 VLAN interfaces with IP addresses and VRRP virtual gateways for VLANs 10, 20, 30, 40, and 50.

We set several access ports in VLAN 20 and multiple trunk ports allowing different combinations of VLANs (10, 20, 30, 40, 50).

We added a static route for 192.168.3.0/24 via 192.168.2.3.

3- SW3

```
<SW7>disp cu
#
sysname SW7
#
vlan batch 10 20 30 40 50 100
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
---- More ----|
```

LSW3

```
#
interface Vlanif1
#
interface Vlanif10
ip address 192.168.3.2 255.255.255.0
vrrp vrid 10 virtual-ip 192.168.3.1
vrrp vrid 10 priority 120
#
interface Vlanif20
ip address 192.168.4.2 255.255.255.0
vrrp vrid 20 virtual-ip 192.168.4.1
vrrp vrid 20 priority 120
#
interface Vlanif30
ip address 10.200.1.2 255.255.255.0
vrrp vrid 30 virtual-ip 10.200.1.1
#
interface Vlanif40
ip address 10.150.1.2 255.255.255.0
vrrp vrid 40 virtual-ip 10.150.1.1
vrrp vrid 40 priority 120
#
interface Vlanif50
ip address 192.168.50.2 255.255.255.0
vrrp vrid 50 virtual-ip 192.168.50.1
vrrp vrid 50 priority 120

Dec  2 2025 11:00:34-08:00 SW7 %%01PHY/1/PHY(1)[0]:   Ethernet0/0/3: change sta
tus to up#
---- More ----
```

```
LSW3
tus to up#
interface MEth0/0/1
#
interface Ethernet0/0/1
port link-type access
port default vlan 50
#
interface Ethernet0/0/2
port link-type access
port default vlan 50
#
interface Ethernet0/0/3
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50
#
interface Ethernet0/0/4
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50
#
interface Ethernet0/0/5
#
interface Ethernet0/0/6
#
interface Ethernet0/0/7

Dec 2 2025 11:01:01-08:00 SW7 %%01CPUUSAGE/4/CPU_USAGE_RESUME(1)[1]:CPU utilization recovered to the normal range.
Dec 2 2025 11:01:01-08:00 SW7 %%01VOSCPU/4/CPU_USAGE_RESUME(1)[2]:CPU utilization recovered to the normal range.
interface Ethernet0/0/8
More
interface GigabitEthernet0/0/1
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50
#
interface GigabitEthernet0/0/2
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50
#
interface NULL0
#
user-interface con 0
user-interface vty 0 4
#
return
```

We created VLANs 10, 20, 30, 40, and 50.

We configured Layer 3 VLAN interfaces with IP addresses and VRRP virtual gateways for VLANs 10, 20, 30, 40, and 50, setting higher VRRP priority (120) for active role.

We set several access ports (Eth0/0/1–2) in VLAN 50 and multiple trunk ports (Eth0/0/3–4, Gi0/0/1–2) allowing VLANs 10, 20, 30, 40, 50.

4- SW4

LSW4

```
<SW5>disp cu
#
sysname SW5
#
vlan batch 10 20 30 40 50
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
interface Vlanif1
#
interface Vlanif10
 ip address 192.168.3.3 255.255.255.0
 ---- More ----|
```

LSW4

```
interface Vlanif1
#
interface Vlanif10
 ip address 192.168.3.3 255.255.255.0
 vrrp vrid 10 virtual-ip 192.168.3.1
#
interface Vlanif20
 ip address 192.168.4.3 255.255.255.0
 vrrp vrid 20 virtual-ip 192.168.4.1
#
interface Vlanif30
 ip address 10.200.1.3 255.255.255.0
 vrrp vrid 30 virtual-ip 10.200.1.1
#
interface Vlanif40
 ip address 10.150.1.3 255.255.255.0
 vrrp vrid 40 virtual-ip 10.150.1.1
#
interface Vlanif50
 ip address 192.168.50.3 255.255.255.0
 vrrp vrid 50 virtual-ip 192.168.50.1
#
interface MEth0/0/1
#
interface Ethernet0/0/1
#
interface Ethernet0/0/2
 port link-type access
 port default vlan 30
 ---- More ----|
```

LSW4

```
#
interface Ethernet0/0/2
  port link-type access
  port default vlan 30
#
interface Ethernet0/0/3
  port link-type access
  port default vlan 30
#
interface Ethernet0/0/4
  port link-type trunk
  port trunk allow-pass vlan 30 40 50
#
interface Ethernet0/0/5
  port link-type trunk
  port trunk allow-pass vlan 10 20 30 40 50
#
```

```
interface GigabitEthernet0/0/1
  port link-type trunk
  port trunk allow-pass vlan 10 20 30 40 50
#
interface GigabitEthernet0/0/2
  port link-type trunk
  port trunk allow-pass vlan 10 20 30 40 50
#
interface NULL0
#
user-interface con 0
user-interface vty 0 4
#
return
```

We created VLANs 10, 20, 30, 40, and 50.

We configured Layer 3 VLAN interfaces with IP addresses and VRRP virtual gateways for VLANs 10, 20, 30, 40, and 50.

We set access ports (Eth0/0/2–3) in VLAN 30 and multiple trunk ports (Eth0/0/4–5, Gi0/0/1–2) allowing VLANs 10, 20, 30, 40, 50.

5- SW5

LSW5

```
<SW4>disp cu
#
sysname SW4
#
vlan batch 10 20 30 40 50
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
interface Vlanif1
#
interface Vlanif10
 ip address 192.168.3.5 255.255.255.0
|
```

LSW5

```
#
interface Vlanif10
 ip address 192.168.3.5 255.255.255.0
 vrrp vrid 10 virtual-ip 192.168.3.1
#
interface Vlanif20
 ip address 192.168.4.5 255.255.255.0
 vrrp vrid 20 virtual-ip 192.168.4.1
#
interface Vlanif30
 ip address 10.200.1.5 255.255.255.0
 vrrp vrid 30 virtual-ip 10.200.1.1
#
interface Vlanif40
 ip address 10.150.1.5 255.255.255.0
 vrrp vrid 40 virtual-ip 10.150.1.1
#
interface Vlanif50
 ip address 192.168.50.5 255.255.255.0
 vrrp vrid 50 virtual-ip 192.168.50.1
#
interface MEth0/0/1
#
interface Ethernet0/0/1
#
interface Ethernet0/0/2
 port link-type access
 port default vlan 40
#
---- More ----|
```

LSW5

```
#
interface Ethernet0/0/2
 port link-type access
 port default vlan 40
#
interface Ethernet0/0/3
 port link-type access
 port default vlan 40
#
interface Ethernet0/0/4
 port link-type access
 port default vlan 40
#
```

```
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk allow-pass vlan 10 20 30 40
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
user-interface con 0
user-interface vty 0 4
#
port-group vlan40
#
return
---- More ----|
```

We created VLANs 10, 20, 30, 40, and 50.

We configured Layer 3 VLAN interfaces with IP addresses and VRRP virtual gateways for VLANs 10, 20, 30, 40, and 50.

We set access ports (Eth0/0/2–4) in VLAN 40 and a trunk port (Gi0/0/1) allowing VLANs 10, 20, 30, 40, 50.

We enabled port-group VLAN 40.

- **Access Controllers**

- 1- AC1**

```
AC3
<AC>disp cu
#
 sysname AC
#
 set memory-usage threshold 0
#
 ssl renegotiation-rate 1
#
 vlan batch 10 20 30 40 50 60 70
#
 authentication-profile name default_authen_profile
 authentication-profile name dot1x_authen_profile
 authentication-profile name mac_authen_profile
 authentication-profile name portal_authen_profile
 authentication-profile name macportal_authen_profile
#
 dhcp enable
#
 diffserv domain default
#
 radius-server template default
#
 pki realm default
   rsa local-key-pair default
   enrollment self-signed
#
 ike proposal default
   encryption-algorithm aes-256
   dh group14
   ---- More ----|
```

AC3

```
dh group14
authentication-algorithm sha2-256
authentication-method pre-share
integrity-algorithm hmac-sha2-256
prf hmac-sha2-256
#
free-rule-template name default_free_rule
#
portal-access-profile name portal_access_profile
#
ip pool pool1
 gateway-list 192.168.250.1
 network 192.168.250.0 mask 255.255.255.0
#
ip pool pool3
 gateway-list 10.200.1.1
 network 10.200.1.0 mask 255.255.255.0
#
aaa
 authentication-scheme default
 authentication-scheme radius
 authentication-mode radius
 authorization-scheme default
 accounting-scheme default
 domain default
 authentication-scheme radius
 radius-server default
 domain default_admin
 authentication-scheme default
---- More ----
```

AC3

```
authentication-scheme default
local-user admin password irreversible-cipher $1a$+@X&Td7JhD$D{VN%(W/OWHK:L'Wx;
|%-NVx/Y1\U28&wBF2pJw/$
local-user admin privilege level 15
local-user admin service-type http
#
interface Vlanif30
 ip address 10.200.1.5 255.255.255.0
 dhcp select global
#
interface Vlanif50
 ip address 192.168.250.1 255.255.255.0
 dhcp select global
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk pvid vlan 50
 port trunk allow-pass vlan 10 20 30 40 50 60
#
interface GigabitEthernet0/0/2
 port link-type trunk
 port trunk allow-pass vlan 10 20 30 40 50 60 70
#
interface GigabitEthernet0/0/3
#
interface GigabitEthernet0/0/4
#
interface GigabitEthernet0/0/5
#
---- More ----
```

AC3

```

interface GigabitEthernet0/0/7
  undo negotiation auto
  duplex half
#
interface GigabitEthernet0/0/8
  undo negotiation auto
  duplex half
#
interface NULL0
#
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
#
ssh server secure-algorithms cipher aes256_ctr aes128_ctr
ssh server key-exchange dh_group14_shal
ssh client secure-algorithms cipher aes256_ctr aes128_ctr
ssh client secure-algorithms hmac sha2_256
ssh client key-exchange dh_group14_shal
#
capwap source interface vlanif50
#
user-interface con 0
  authentication-mode password
user-interface vty 0 4
  protocol inbound all
user-interface vty 16 20
  protocol inbound all
#
wlan
  ---- More ----|

```

AC3

```

#
wlan
  traffic-profile name default
  security-profile name sec
    security wpa-wpa2 psk pass-phrase %^%#q[PiW:d'E%p]0\Dkd}N=Gu2Z$\b;e93c@w+I`v*Q
    %^%# aes
  security-profile name default
  security-profile name default-wds
  security-profile name default-mesh
  ssid-profile name ssid
    ssid logy
  ssid-profile name default
  vap-profile name vap
    forward-mode tunnel
    service-vlan vlan-id 30
    ssid-profile ssid
    security-profile sec
  vap-profile name default
  wds-profile name default
  mesh-handover-profile name default
  mesh-profile name default
  regulatory-domain-profile name reg
    country-code EG
  regulatory-domain-profile name default
  air-scan-profile name default
  rrm-profile name default
  radio-2g-profile name default
  radio-5g-profile name default
  wids-spoof-profile name default
  ---- More ----|

```

AC3

```
radio-2g-profile name default
radio-5g-profile name default
wids-spoof-profile name default
wids-profile name default
wireless-access-specification
ap-system-profile name default
port-link-profile name default
wired-port-profile name default
serial-profile name preset-enjoyor-toeap
ap-group name lab
    regulatory-domain-profile reg
radio 0
    vap-profile vap wlan 1
radio 1
    vap-profile vap wlan 1
radio 2
    vap-profile vap wlan 1
ap-group name default
ap-id0 type-id 69 ap-mac 00e0-fc92-4610 ap-sn 210235448310614B0626
    ap-name logy
    ap-group lab
provision-ap
#
dot1x-access-profile name dot1x_access_profile
#
mac-access-profile name mac_access_profile
#
return
<AC>
<AC>|
```

- 1- VLANs: 10, 20, 30, 40, 50, 60, 70 created.
- 2- IP Pools / DHCP:
 - pool1: 192.168.250.0/24, gateway 192.168.250.1
 - pool3: 10.200.1.0/24, gateway 10.200.1.1
- 3- Interfaces:
 - Vlanif30 → IP 10.200.1.5, DHCP global
 - Vlanif50 → IP 192.168.250.1, DHCP global
 - Gi0/0/1 & Gi0/0/2 → trunk ports passing multiple VLANs
 - Gi0/0/7 & Gi0/0/8 → half-duplex

4- WLAN / Wireless:

- WLAN named (logy) with SSID profile, VAP profile, WPA/WPA2 PSK security
- Radios configured for VAP (vap) on VLAN 30
- AP group (lab) provisioned with AP (logy)

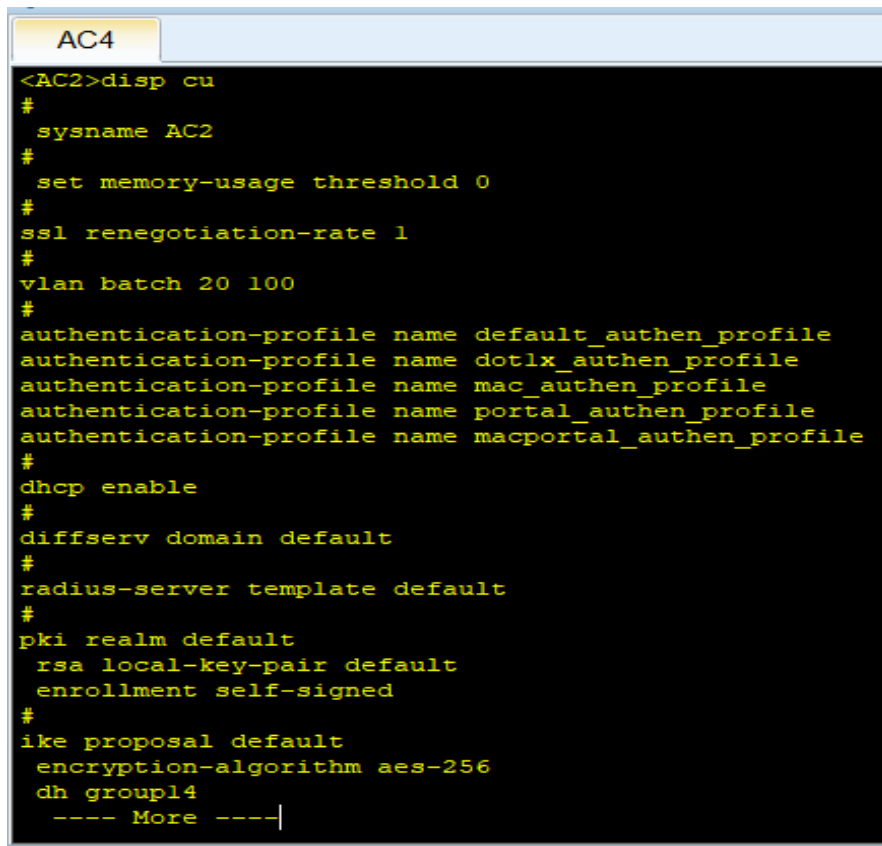
5- Authentication / AAA:

Local admin user with HTTP access

6- Management / Security:

CAPWAP source interface: Vlanif50

2- AC2



```
<AC2>disp cu
#
 sysname AC2
#
 set memory-usage threshold 0
#
 ssl renegotiation-rate 1
#
 vlan batch 20 100
#
 authentication-profile name default_authen_profile
 authentication-profile name dot1x_authen_profile
 authentication-profile name mac_authen_profile
 authentication-profile name portal_authen_profile
 authentication-profile name macportal_authen_profile
#
 dhcp enable
#
 diffserv domain default
#
 radius-server template default
#
 pki realm default
   rsa local-key-pair default
   enrollment self-signed
#
 ike proposal default
 encryption-algorithm aes-256
 dh group14
 ---- More ----
```

AC4

```

dh group14
authentication-algorithm sha2-256
authentication-method pre-share
integrity-algorithm hmac-sha2-256
prf hmac-sha2-256
#
free-rule-template name default_free_rule
#
portal-access-profile name portal_access_profile
#
ip pool pool-mgm
gateway-list 192.168.5.1
network 192.168.5.0 mask 255.255.255.0
#
ip pool pool-sta
gateway-list 192.168.4.1
network 192.168.4.0 mask 255.255.255.0
#
aaa
authentication-scheme default
authentication-scheme radius
authentication-mode radius
authorization-scheme default
accounting-scheme default
domain default
authentication-scheme radius
radius-server default
domain default_admin
authentication-scheme default
---- More ----

```

AC4

```

authentication-scheme default
local-user admin password irreversible-cipher $1a$qFj''MKOm+$EAIp*d,b(S'\vA%IX<
oAod@^A.TR|L'us`:bGk7W$
local-user admin privilege level 15
local-user admin service-type http
#
interface Vlanif20
ip address 192.168.4.3 255.255.255.0
dhcp select global
#
interface Vlanif100
ip address 192.168.5.2 255.255.255.0
dhcp select global
#
interface GigabitEthernet0/0/1
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 20 100
#
interface GigabitEthernet0/0/2
port link-type trunk
port trunk allow-pass vlan 20 100
#
interface GigabitEthernet0/0/3
#
interface GigabitEthernet0/0/4
#
interface GigabitEthernet0/0/5
#
---- More ----

```

AC4

```
interface GigabitEthernet0/0/7
  undo negotiation auto
  duplex half
#
interface GigabitEthernet0/0/8
  undo negotiation auto
  duplex half
#
interface NULL0
#
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
#
ssh server secure-algorithms cipher aes256_ctr aes128_ctr
ssh server key-exchange dh_group14_shal
ssh client secure-algorithms cipher aes256_ctr aes128_ctr
ssh client secure-algorithms hmac sha2_256
ssh client key-exchange dh_group14_shal
#
capwap source interface vlanif100
#
user-interface con 0
  authentication-mode password
user-interface vty 0 4
  protocol inbound all
user-interface vty 16 20
  protocol inbound all
#
wlan
---- More ----|
```


AC4

```
wlan
traffic-profile name default
security-profile name sec
  security wpa-wpa2 psk pass-phrase %^%#{4_LU>n/,Thsu|<rznl(>BU5$NHdP5KNw\8E9:9D
%^%# aes
security-profile name default
security-profile name default-wds
security-profile name default-mesh
ssid-profile name ssid
  ssid logy
ssid-profile name default
vap-profile name vap
  forward-mode tunnel
  service-vlan vlan-id 20
  ssid-profile ssid
  security-profile sec
vap-profile name default
wds-profile name default
mesh-handover-profile name default
mesh-profile name default
regulatory-domain-profile name yasmen
  country-code EG
regulatory-domain-profile name default
air-scan-profile name default
rrm-profile name default
radio-2g-profile name default
radio-5g-profile name default
wids-spoof-profile name default
wids-profile name default
---- More ----|
```

AC4

```
wids-spoof-profile name default
wids-profile name default
wireless-access-specification
ap-system-profile name default
port-link-profile name default
wired-port-profile name default
serial-profile name preset-enjoyor-toeap
ap-group name yasmen
  regulatory-domain-profile yasmen
  radio 0
    vap-profile vap wlan 1
  radio 1
    vap-profile vap wlan 1
  radio 2
    vap-profile vap wlan 1
ap-group name default
ap-id 0 type-id 69 ap-mac 00e0-fc1e-68a0 ap-sn 2102354483103F6D004E
  ap-name yasmen
  ap-group yasmen
provision-ap
#
dot1x-access-profile name dot1x_access_profile
#
mac-access-profile name mac_access_profile
#
```

- Created VLANs 20 and 100
- Configured DHCP pools for management (192.168.5.0/24) and station clients (192.168.4.0/24)
- Assigned IPs:
 - Vlanif20 → 192.168.4.3
 - Vlanif100 → 192.168.5.2
- Enabled DHCP and used it on both VLAN interfaces
- Configured trunk ports (Gi0/0/1 & Gi0/0/2) allowing VLANs 20 and 100
- Set CAPWAP source to Vlanif100
- Enabled SNMP, SSH security algorithms, and AAA
- Configured WLAN service:
 - SSID logy
 - WPA/WPA2 PSK
 - Forward mode: tunnel
 - Service VLAN: 20
- Provisioned 1 AP (name yasmen) under AP group yasmen with regulatory domain EG
- Configured radios 0, 1, and 2 to broadcast the VAP